

## **Requirement**

**Low power data logging**

**Read RTC , TEMP sensor and aquire data**

**Store data on to flash memory as a CSV files using LittleFs File System**

**Flash Waeear Leaveing**

**Indicator LED Functions -> Wakeup Indicate , Config Mode**

**Battery Voltage analo Reading**

**Download Mode , whan button press at boot**

**View Logged data from Flash memeory. (Show information Size,Raw Count)**

**Delete a file and folder from the storage.**

**Download Logged data from Flash memeory to Computer via USB.**

**Set Device parameters**

- Set RTC Date**
- Set RTC Time**
- Sync RTC Date Time From PC**
- Set data log interval**
- Set heartbeat STATUS LED**

**View Device Parameters**

- View RTC Date**
- View RTC Time**
- View data log interval**
- View Battery Voltage**
- View Sensor Readings**
- View Avilble Storage**
- View heartbeat STATUS LED configs**

**Restart Logger via USb Serial commands.**

**Format Storage. (Password Protected)**

**Download all the data files once.**

**RTC Fail Safe Logging mechanism.**

**External Flash storage usage if internal got full. Also data retrieval from external flash storage**

**UART based data transfer and config**

**Bluetooth based wireless config and data transfer**

**Python based GUI Config data , analyze data on PC**

**Status LED Heartbeat Feature**

Data Analyz tool

Temp,Hume,Vabt vs Time

Reduce data points interval (Down Sample)

Zoom to specific time interval

plot moving avg curve (N-Point)

Find Min,MAX,AVG

Remove outliers

**Developed**

**Tested**

Yes

Yes

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|-----------|-----------|
| No        | No        |
| Yes       | Yes       |
| No        | No        |
| Yes       | Yes       |
| No        | No        |
| Partially | Partially |
| Yes       | Yes       |

## Comments

Sleep Mode Current : 25 uA

Active mode avg Current : 8.05mA

Automatically handle by littleFs File System.

Tested and calibrated in range (3.3V to 4.3V)